MIL-M-38510/383A
AMENDMENT 2
9 January 1996
SUPERSEDING
AMENDMENT 1
28 July 1992

MILITARY SPECIFICATION

Inactive for new design as of 8 January 1996

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED LOW-POWER SCHOTTKY TTL,
OCTAL BUFFER GATES WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/383A, dated 11 February 1988 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 15

TABLE III, device type 01, truth table tests, test number 101, pin 19: Delete "A" and substitute "B".

PAGE 17

TABLE III, device type 01, footnote 3/, delete in its entirety and substitute: "3/ I_O limits shall be -20 mA min/-112 mA max for circuit A and -30 mA min/-110 mA max for circuit C".

PAGE 19

TABLE III, device types 02 and 03, subgroup 2, delete in its entirety and substitute: "Same tests, terminal conditions and limits as subgroup 1 except T_C = +125°C, V_{IC} tests are omitted, and V_{IL} = +0.7 V."

PAGE 20

TABLE III, device types 02 and 03, t_{PLH1}, test numbers 102 through 109, maximum limits column: Delete "13/" and substitute "11".

TABLE III, device types 02 and 03, t_{PHL1}, test numbers 110 through 117, maximum limits column: Delete "10" and substitute "12".

TABLE III, device types 02 and 03, t_{PHL1}, test numbers 114 and 115, pin 19: Add "7/".

TABLE III, device types 02 and 03, t_{PZL}, test numbers 118 through 125, minimum and maximum limits columns: Delete "7" and "14/" and substitute "13/" and "15", respectively.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
* 3	3
* 4	4

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

PAGE 20-Continued

TABLE III, device types 02 and 03, t_{PZH}, test numbers 126 through 133, minimum and maximum limits columns: Delete "7" and "15/" and substitute "5" and "15", respectively.

TABLE III, device types 02 and 03, t_{PLZ}, test numbers 134 through 141, minimum and maximum limits columns: Delete "3" and "" and substitute "2" and "10", respectively.

*TABLE III, device type 03 only, t_{PLH1}, delete "3 ns min" and substitute "2 ns min".
*TABLE III, device type 03 only, t_{PZH}, delete "5 ns min" and substitute "2 ns min".
*TABLE III, device type 03 only, t_{PZL}, delete "13/ min" and substitute "2 ns min".
*TABLE III, device type 03 only, t_{PLZ}, delete "2 ns min" and substitute "1 ns min".

PAGE 21

TABLE III, device types 02 and 03, t_{PLH1}, maximum limits column: Delete "16/" and substitute "13".

TABLE III, device types 02 and 03, t_{PHL1}, maximum limits column: Delete "13" and substitute "14/".

TABLE III, device types 02 and 03, t_{PZL}, minimum and maximum limits columns: Delete "7" and "25" and substitute "13/" and "18", respectively.

TABLE III, device types 02 and 03, t_{PZH}, minimum and maximum limits columns: Delete "7" and "25" and substitute "5" and "18", respectively.

TABLE III, device types 02 and 03, t_{PLZ}, minimum and maximum limits columns: Delete "3" and "17/" and substitute "2" and "12", respectively.

TABLE III, device types 02 and 03, footnote 6/, delete in its entirety and substitute: "6/ I_O limits shall be -20 mA min/-112 mA max for circuit A and -30 mA min/-110 mA max for circuit C".

TABLE III, device types 02 and 03, footnote 7/, delete in its entirety and substitute: "7/ Apply 3.0 V min/5.5 V max for device type 02; apply GND for device type 03."

*TABLE III, footnote 9/, device type 03 delete "15" and substitute "18", delete "24" and substitute "25", delete "27" and substitute "29".

TABLE III, device types 02 and 03, footnote 13/, delete in its entirety and substitute: "13/ 5 ns for device type 02 and 4 ns for device type 03."

TABLE III, device types 02 and 03, footnote 14/, delete in its entirety and substitute: "14/ 14 ns for device type 02 and 16 ns for device type 03."

TABLE III, device types 02 and 03, footnote 15/: Delete in its entirety.

TABLE III, device types 02 and 03, footnote 16/: Delete in its entirety.

TABLE III, device types 02 and 03, footnote 17/: Delete in its entirety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last prvious amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1590)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified		Device type	Limits		Unit
					Min	Max	
High level output voltage	V_{OH1}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, I_{OH} = -3 mA V_{IH} = 2.0 V (02, 03 only) T_C = 125°C, V_{IL} = 0.7 V		All	2.4		V
High level output voltage	V_{OH2}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, I_{OH} = -12 mA V_{IH} = 2.0 V (02, 03 only) T_C = 125°C, V_{IL} = 0.7 V		All	2.0		V
Low level output voltage	V_{OL1}	V_{CC} = 4.5 V, I_{OL} = 12 mA, V_{IH} = 2.0 V, V_{IL} = 0.8 V T_C = 125°C, V_{IL} = 0.7 V		All		0.4	V
Low level output voltage	V_{OL2}	V_{CC} = 4.5 V, I_{OL} = 18 mA, V_{IH} = 2.0 V, V_{IL} = 0.8 V T_C = 125°C, V_{IL} = 0.7 V		All		0.45	V
Off-state (high-impedance state) output current	I_{OZH}	V_{CC} = 5.5 V, V_{OUT} = 2.7 V, V_{IH} = 2.0 V		All		20	µA
Off-state (low-impedance state) output current	I_{OZL}	V_{CC} = 5.5 V, V_{OUT} = 0.4 V, V_{IH} = 2.0 V		All		-20	µA
Low level input current (all inputs)	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		All	0	-100	µA
High level input current (all inputs)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		All		20	µA
High level input current (all inputs)	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V		All		100	µA
Output current 1/	I_O	V_{CC} = 5.5 V, V_{OUT} = 2.25 V		All	-20	-112	mA
*Supply current	I_{CCH}	V_{CC} = 5.5 V	Outputs high	01		10	mA
				02		17	
				03		18	
	I_{CCL}		Outputs low	01		23	
				02		28	
				03		25	
	I_{CCZ}		All outputs disabled	01		25	
				02		32	
				03		29	

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C	All		-1.5	V
*Propagation delay time (low-to-high level)	t_{PLH1}	C_L = 50 pF, R_L = 500Ω, V_{CC} = 5.0 V	01	3	12	ns
			02	3	13	
			03	2	13	
Propagation delay time (high-to-low level)	t_{PHL1}		01	2	11	ns
			02	3	14	
			03	3	16	
*Output enable time to low level	t_{PZL}		01	5	20	ns
			02	5	18	
			03	2	18	
*Output enable time to high level	t_{PZH}		01	5	15	ns
			02	5	18	
			03	2	18	
*Output disable time from low level	t_{PLZ}		01	3	18	ns
			02	2	12	
			03	1	12	
Output disable time from high level	t_{PHZ}		01,02, 03	2	12	ns

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.